

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA1017	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
Student Name:	LOSHINI.L	Reg. No.:	192571015
Date:	18/8/26	Duration:	60 minutes

Code of Conduct

I LOSHINI.L 192571015 _ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: LOSHINI.L_____

Annexure A – Test Case Design Assignment

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **design and document comprehensive test cases** to verify the correctness, functionality, reliability, and usability of the proposed system.

Your test case design should include:

1. Identify the **functional and non-functional requirements** to be tested.
2. Identify the **test scenarios** for the assigned problem statement.
3. Prepare detailed **test cases** covering normal, boundary, invalid, and exceptional conditions.
4. Include **Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status**.
5. Apply appropriate **test design techniques** such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.
6. Ensure that the test cases provide adequate **requirement and functional coverage** for the assigned system.

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	Mark
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4

Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25

Problem Statement

Improper segregation of municipal waste reduces recycling efficiency and increases landfill usage. Existing waste processing facilities often rely on manual sorting, leading to inconsistent results and higher operational costs. An AI-powered waste segregation system should identify different categories of waste using computer vision, automate sorting through robotic mechanisms, monitor recycling statistics, and generate waste management reports. The platform should improve recycling efficiency while reducing environmental pollution.

1. Requirement Identification

Before designing test cases, the functional and non-functional requirements of the AI-Based Smart Waste Segregation and Recycling Automation System were identified. These requirements form the basis for test scenario and test case design.

1.1 Functional Requirements (FR)

ID	Functional Requirement
FR1	The system shall capture an image/video frame of each waste item on the conveyor belt via camera.
FR2	The system shall classify each waste item into one of the defined categories (Plastic, Paper, Metal, Glass, Organic, E-waste) using a computer vision model.
FR3	The system shall trigger the robotic actuator to route the classified item into the correct bin.
FR4	The system shall log every sorting event with timestamp, category, and confidence score.
FR5	The system shall compute and display recycling statistics (daily/weekly/monthly) on a dashboard.
FR6	The system shall generate downloadable waste management reports (PDF/Excel) for a selected date range.
FR7	The system shall allow authorized facility managers to log in and view the performance dashboard.
FR8	The system shall send alert notifications for misclassifications, low-confidence items, or actuator faults.
FR9	The system shall allow an operator to manually override an automated sorting decision.
FR10	The system shall support multiple simultaneous conveyor lines/facilities under one platform.

1.2 Non-Functional Requirements (NFR)

ID	Non-Functional Requirement
NFR1	Classification response time shall be less than or equal to 500 ms per item to match real-time conveyor sorting speed.
NFR2	The system shall maintain at least 99% uptime during facility operating hours.
NFR3	The computer vision model shall achieve at least 95% classification accuracy (precision/recall) per waste category.
NFR4	The platform shall scale to support multiple conveyor lines and facilities without performance degradation.
NFR5	All sorting data and user credentials shall be encrypted in storage and in transit.
NFR6	The dashboard shall be usable by non-technical facility staff with minimal training.

ID	Non-Functional Requirement
NFR7	The robotic actuation subsystem shall include a fail-safe stop mechanism to prevent unsafe operation.
NFR8	Waste management reports for a one-month period shall be generated within 5 seconds.

2. Test Scenario Identification

Based on the requirements above, the following high-level test scenarios were identified. Each scenario groups related test cases and is later mapped to requirements in the traceability matrix (Section 6).

TS ID	Test Scenario	Linked Requirement(s)
TS01	Verify waste image classification accuracy for each valid category.	FR2, NFR3
TS02	Verify robotic sorting actuation matches the classification result.	FR3
TS03	Verify system behavior for low-confidence / ambiguous classification.	FR2, FR8, NFR3
TS04	Verify logging of every sorting event.	FR4
TS05	Verify recycling statistics calculation and dashboard update.	FR5
TS06	Verify waste management report generation and export.	FR6, NFR8
TS07	Verify facility manager login and access control.	FR7, NFR5
TS08	Verify alert notification on actuator or classification faults.	FR8
TS09	Verify manual override of an automated sorting decision.	FR9
TS10	Verify system handling of multiple/overlapping items on the conveyor.	FR2, FR3
TS11	Verify system recovery after camera or sensor failure.	NFR2, NFR7
TS12	Verify system performance under high item throughput (load).	NFR1, NFR4

2.1 Test Scenario Category Mindmap

The mindmap organizes the testing scenarios into **six major focus areas** to ensure the complete system is tested effectively:

- **Classification Testing** – Checks waste category accuracy, confidence levels, and invalid images.
- **Robotic Sorting Testing** – Verifies actuator movement, bin routing, and safe-stop operation.
- **Alerts & Notification Testing** – Tests fault alerts, low-confidence warnings, and escalation.
- **Performance & Reliability Testing** – Measures system speed, response time, uptime, and recovery.
- **Data & Reporting Testing** – Ensures accurate statistics, reports, dates, and exported data.
- **Access & Security Testing** – Validates login, account security, user roles, and manual override.



Figure A: Test Scenario Category Mindmap

3. Detailed Test Cases

The following test cases cover normal, boundary, invalid, and exceptional conditions, derived using Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition techniques (see Section 4). Actual Result and Status are left for the tester to complete during execution.

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC01	TS01 – Normal	1. Place a clean, well-lit plastic bottle on the conveyor. 2. Allow camera to capture image. 3. Observe classification output.	Image: PET plastic bottle, clear background	Item classified as "Plastic" with confidence $\geq 95\%$; routed to Plastic bin.	Pending execution	Pending
TC02	TS01 – Normal	1. Place a paper sheet/cardboard piece on the conveyor. 2. Capture image. 3. Observe classification.	Image: Paper/cardboard item	Item classified as "Paper" with confidence $\geq 95\%$; routed to Paper bin.	Pending execution	Pending
TC03	TS01 – Normal	1. Place an aluminium can on the conveyor. 2. Capture image. 3. Observe classification.	Image: Metal can	Item classified as "Metal" with confidence $\geq 95\%$; routed to Metal bin.	Pending execution	Pending
TC04	TS01 – Normal	1. Place a glass bottle on the conveyor. 2. Capture image. 3. Observe classification.	Image: Glass bottle	Item classified as "Glass" with confidence $\geq 95\%$; routed to Glass bin.	Pending execution	Pending
TC05	TS01 – Normal	1. Place food scraps on the conveyor. 2. Capture image. 3. Observe classification.	Image: Organic food waste	Item classified as "Organic" with confidence $\geq 95\%$; routed to Organic bin.	Pending execution	Pending
TC06	TS01 – Normal	1. Place a small battery/electronic item on the conveyor. 2. Capture image. 3. Observe classification.	Image: Small battery	Item classified as "E-waste" with confidence $\geq 95\%$; routed to E-waste bin.	Pending execution	Pending

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC07	TS03 – Boundary (BVA)	1. Feed an item whose model output confidence is exactly at the threshold. 2. Observe system decision.	Confidence score = 95.0% (threshold value)	Item is accepted as a valid auto-classification (boundary inclusive) and routed normally.	Pending execution	Pending

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC08	TS03 – Boundary (BVA)	1. Feed an item whose model output confidence is just below the threshold. 2. Observe system decision.	Confidence score = 94.9%	Item is flagged "Low Confidence" and diverted to the manual review lane instead of auto-sorting.	Pending execution	Pending
TC09	TS03 – Invalid (EP)	1. Feed a blurry / unrecognizable image. 2. Observe system decision.	Image: heavily blurred / obstructed item	System returns "Unclassified"; item diverted to manual sorting lane; event logged.	Pending execution	Pending
TC10	TS11 – Exceptional	1. Disconnect/simulate camera failure while conveyor is running. 2. Observe system behavior.	Camera feed = no signal	System raises a "Camera Error" alert and halts the conveyor safely (fail-safe stop, NFR7).	Pending execution	Pending
TC11	TS10 – Exceptional	1. Place two waste items overlapping in a single camera frame. 2. Observe classification behavior.	Image: two overlapping items (e.g., plastic + paper)	System detects multiple objects and either classifies each separately or flags the frame for manual sorting.	Pending execution	Pending
TC12	TS02 / TS08 – Exceptional	1. Simulate the robotic actuator failing to move to the assigned bin. 2. Observe system response.	Simulated actuator fault signal	System detects the actuation failure via sensor feedback, logs the fault, and sends an alert notification (TS08).	Pending execution	Pending
TC13	TS04 / TS05 – Normal	1. Perform 10 sorting events across different categories. 2. Open the statistics dashboard. 3. Compare displayed counts with actual events.	10 mixed-category sorting events	Dashboard statistics reflect the exact count per category in real time.	Pending execution	Pending
TC14	TS06 – Normal	1. Select a valid date range with existing data. 2. Click "Generate Report". 3. Measure generation time and check contents.	Date range: 1 month with sorting data	Report (PDF/Excel) is generated within 5 seconds (NFR8) and contains accurate category totals and contamination rate.	Pending execution	Pending

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC15	TS06 – Boundary (BVA)	1. Select a date range with no sorting data. 2. Click "Generate Report".	Date range with zero events	System displays "No data available" instead of generating an empty or broken report.	Pending execution	Pending
TC16	TS06 – Invalid (EP)	1. Enter an end date earlier than the start date. 2. Click "Generate Report".	Start date = 2026-08-10, End date = 2026-08-01	System shows validation error "End date must be after start date"; report is not generated.	Pending execution	Pending
TC17	TS07 – Normal	1. Enter valid facility-manager username and password. 2. Click "Login".	Valid registered credentials	User is authenticated and redirected to the performance dashboard.	Pending execution	Pending
TC18	TS07 – Boundary (Decision Table)	1. Enter an incorrect password 3 times in a row. 2. Attempt a 4th login.	3 consecutive invalid password attempts	Account is locked after the 3rd failed attempt (boundary value); 4th attempt is rejected with a lockout message.	Pending execution	Pending
TC19	TS09 – Normal	1. Select an item flagged for manual review. 2. Manually assign the correct category. 3. Confirm override.	Flagged item + operator-selected category = "Plastic"	Item is routed to the correct bin; override is logged with operator ID and timestamp.	Pending execution	Pending
TC20	TS12 –	1. Feed items at a sustained rate	100 items/minute	Classification latency stays ≤500	Pending	Pending

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
	Exceptional (Load)	of 100 items/minute. 2. Monitor classification latency and routing accuracy.	mixed-category stream	ms per item (NFR1); no items missed or misrouted.	execution	
TC21	State Transition	1. Start conveyor from Idle. 2. Trigger a fault mid-sort. 3. Reset the fault. 4. Verify conveyor resumes.	Simulated actuator fault, then reset command	State moves Idle → Classifying → Sorting → Fault → Idle correctly; no items are lost or duplicated during the pause.	Pending execution	Pending

4. Application of Test Design Techniques

Multiple test design techniques were applied to ensure systematic and efficient coverage.

4.1 Equivalence Partitioning

Inputs were partitioned into valid and invalid classes so that one representative test per partition gives confidence for the whole class:

- Confidence score: Valid partition ($\geq 95\%$) vs. Invalid partition ($< 95\%$) — TC01–TC06 vs. TC08–TC09.
- Waste category: Six valid categories (Plastic, Paper, Metal, Glass, Organic, E-waste) vs. an invalid/unclassifiable partition — TC01–TC06 vs. TC09.
- Report date range: Valid range with data vs. valid range with no data vs. invalid range (end before start) — TC14, TC15, TC16.

4.2 Boundary Value Analysis (BVA)

Boundary values were tested just at, just below, and just above critical thresholds:

- Classification confidence threshold: 94.9% (just below), 95.0% (exact boundary) — TC07, TC08.
- Login attempt limit: 2nd attempt (still allowed), 3rd attempt (boundary — triggers lockout), 4th attempt (locked out) — TC18.
- Report date range: zero-length/empty data range as a lower boundary — TC15.

4.3 Decision Table Testing

A decision table was used to determine the correct system action based on combinations of classification confidence, item count, and camera status:

Rule	Confidence $\geq 95\%$?	Single Item Detected?	Camera OK?	Resulting Action
R1	Yes	Yes	Yes	Auto-sort to classified bin (TC01–TC06)
R2	No	Yes	Yes	Route to manual review lane (TC08)
R3	Yes / No	No (multiple items)	Yes	Flag frame for manual sorting (TC11)
R4	—	—	No	Halt conveyor and raise camera-error alert (TC10)

4.4 State Transition Testing

The conveyor/sorting subsystem was modeled as a state machine with states Idle, Classifying, Sorting, Manual Review, and Fault/Paused. Test case TC21 exercises the key transitions, including the fault-and-reset cycle, to confirm no items are lost or duplicated. The diagram below illustrates the modeled transitions.

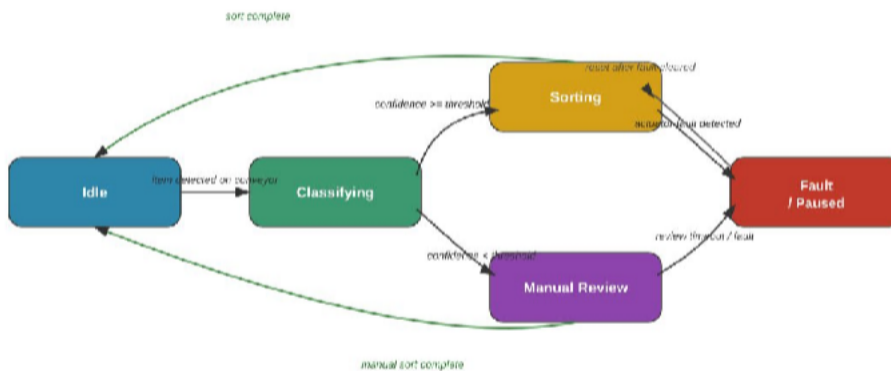


Figure B: State Transition Diagram — Conveyor / Sorting Subsystem

5. Test Coverage Summary

Condition Type	Test Cases	Count
Normal (happy path)	TC01–TC06, TC13, TC14, TC17, TC19	10
Boundary	TC07, TC08, TC15, TC18	4
Invalid	TC09, TC16	2
Exceptional	TC10, TC11, TC12, TC20	4
State Transition	TC21	1

6. Requirement Traceability Matrix

The matrix below maps each requirement to its test scenario(s) and corresponding test case(s), confirming that every functional and non-functional requirement has test coverage.

Requirement	Test Scenario(s)	Test Case(s)
FR1 – Capture image/video	TS01	TC01–TC06
FR2 – Classify waste category	TS01, TS03, TS10	TC01–TC09, TC11
FR3 – Robotic sorting actuation	TS02, TS10	TC01–TC06, TC11, TC12
FR4 – Log sorting events	TS04	TC13
FR5 – Recycling statistics	TS05	TC13
FR6 – Generate reports	TS06	TC14, TC15, TC16
FR7 – Manager login	TS07	TC17, TC18
FR8 – Alerts on faults	TS03, TS08	TC08, TC10, TC12

Requirement	Test Scenario(s)	Test Case(s)
FR9 – Manual override	TS09	TC19
FR10 – Multi-facility support	TS12	TC20
NFR1 – Response time ≤ 500 ms	TS12	TC20
NFR2 – 99% uptime	TS11	TC10, TC21
NFR3 – $\geq 95\%$ classification accuracy	TS01, TS03	TC01–TC08
NFR4 – Scalability	TS12	TC20
NFR5 – Data security	TS07	TC17, TC18
NFR7 – Fail-safe actuation	TS11	TC10, TC12, TC21
NFR8 – Report generation ≤ 5 s	TS06	TC14

7. Conclusion

- The testing process for the **AI-Based Smart Waste Segregation and Recycling Automation System** provides a detailed and structured way to evaluate the complete system before real-world deployment. It covers all major components, including **AI-based waste classification, robotic sorting, conveyor operation, alerts and notifications, data reporting, access control, performance, and fault recovery**. Classification testing ensures that waste items are identified correctly, while robotic sorting testing verifies that the detected waste is directed to the appropriate recycling bin. Performance and reliability testing further ensures that the system can process waste continuously with acceptable response time and minimal failures.
- The use of different testing techniques improves the overall quality and effectiveness of the testing process. **Equivalence Partitioning** checks different groups of valid and invalid inputs, while **Boundary Value Analysis** focuses on important limits such as confidence thresholds and system capacity. **Decision Table Testing** verifies that the correct sorting action is performed for different combinations of conditions. **State Transition Testing** checks the conveyor and robotic system during normal operation, fault conditions, and recovery. Together, these techniques help detect functional errors, unexpected conditions, and safety issues at an early stage.
- The **traceability matrix** ensures that every functional and non-functional requirement is properly connected to one or more test cases, providing complete requirement coverage. Successful execution of these test cases increases confidence in the system's **accuracy, reliability, security, performance, safety, and usability**. Therefore, the proposed testing approach plays an important role in ensuring that the smart waste segregation system is capable of operating efficiently and consistently in a real-world environment, ultimately supporting **automated waste management, improved recycling, reduced human effort, and a cleaner environment**.